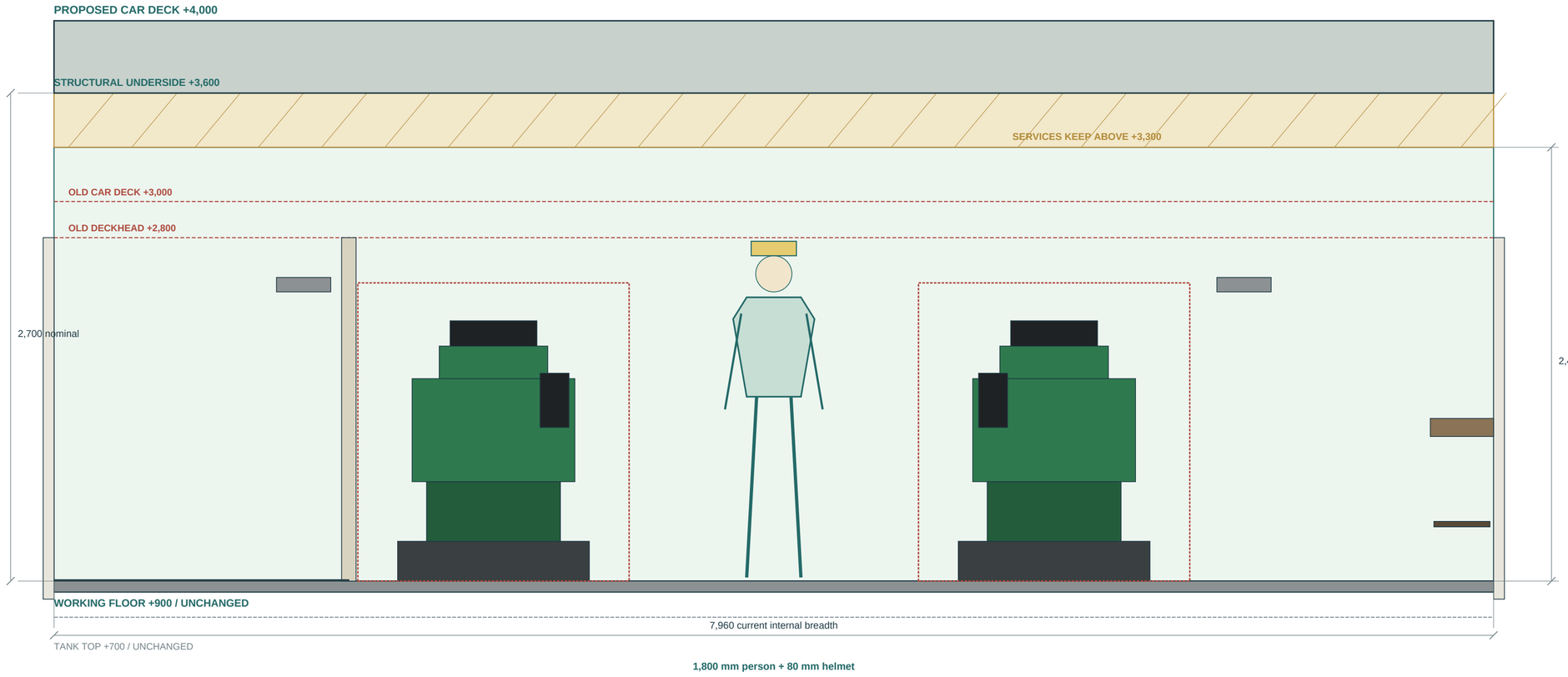
2,400 mm clear below services. Machinery stays at its present level.

The room becomes 800 mm taller at the structural underside. A separate 400 mm vehicle-deck structural reservation makes the proposed car deck 1,000 mm higher. Raising the decks above by the same 1,000 mm retains the existing vehicle and accommodation clearances.



No ducking in designated standing and working areas. Pipes, trays, valves, handles and insulation must stay outside the clear-height envelope.

Engine and gear centres, foundations, tanks and machinery count remain fixed in this concept. Engine top/removal envelopes still require actual supplier data.

400 mm is a structural planning reservation, not a calculated beam size. If selected structure or services need more depth, increase deck elevations; do not consume the 2,400 mm clear space.

What this changes

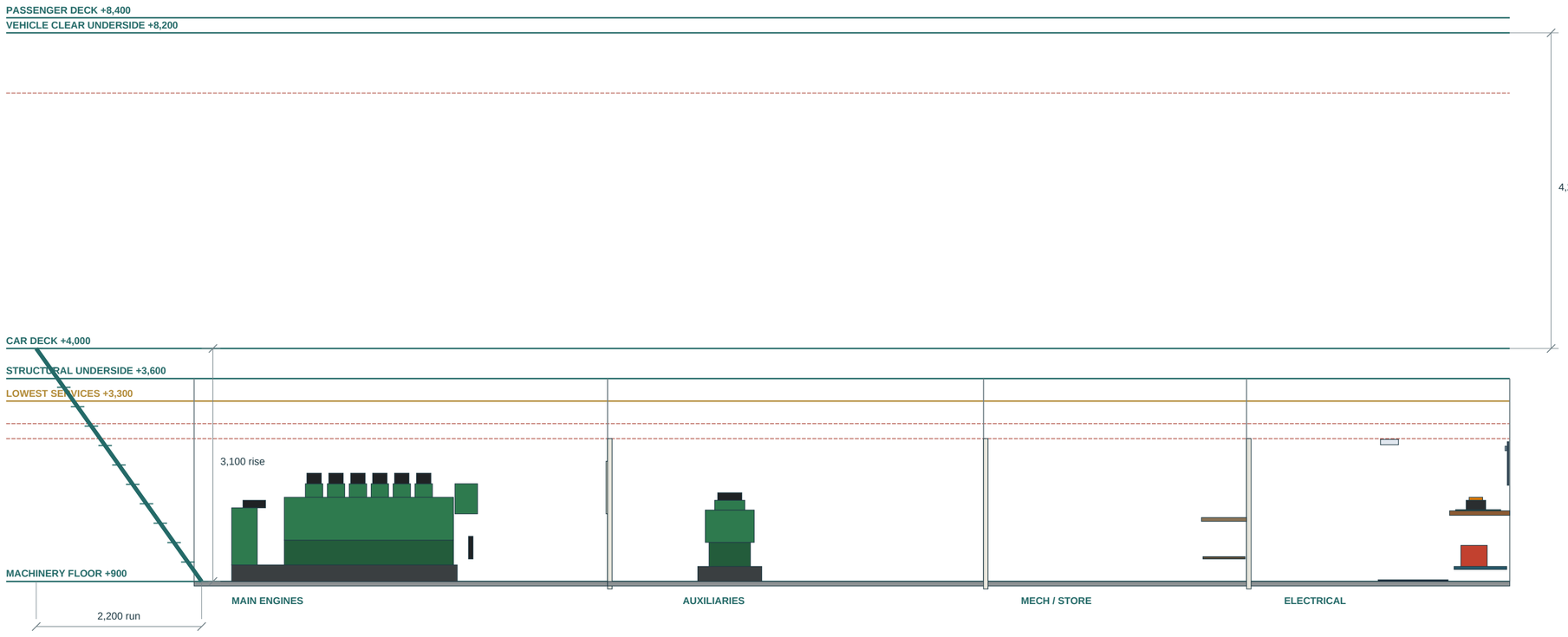
A uniform raised vehicle deck is used to avoid a local machinery hump in the car lanes. Passenger and bridge decks, casings, access, exhausts, vents, ramp hinge/landing arrangements, gangway landings and vertical services need to follow the new levels. Hull breadth, machinery floor and shaft lines are retained as design constraints.

This is the headroom design direction, subject to hull/structure/stability and access coordination. It does not approve a real-vessel conversion. The following ten survey sheets show the existing 1,900 mm baseline for comparison; H1 controls the proposed vertical dimensions.

Deck-height coordination / new ladder and protected working space

REV C | 1:50 section / schedule NTS

Longitudinal coordination / existing machinery remains in place



Proposed port ladder: 800 mm clear; 54.6 deg nominal; head X=6,400, foot X=8,600.

Hatch reservation X=6,400-9,000; retained nominal width 980. Landings, rails, treads and hull clearance require detail.

Datum / function	Existing	H1 proposed	Coordination requirement
Tank top / floor	700 / 900	700 / 900	Retained; no tank or shaft-line lowering.
ER structural underside	2,800	3,600	800 mm gain; keep all services above 3,300 in work areas.
Car / passenger deck	3,000 / 7,400	4,000 / 8,400	Both +1,000; retain 4,200 vehicle clear height.
Bridge / compass deck	10,000 / 12,400	11,000 / 13,400	Raise upper geometry and services consistently.
Main access ladder	2,100 rise	3,100 rise	Extend run 1,600 to 2,200; no steeper shortcut.

01 Vertical design decision

Raise car deck and upper decks by 1,000; raise lowest machinery deck structure by 800. The extra 200 increases the former 200 mm deck envelope to a 400 mm reservation.

02 No silent loss of capacity

Preserve vehicle clear height. Revised overall height, freeboard, lightship/KG, windage, ramp slope, berthing and structure require verification before any game build.

03 Ladder follows deck

A 3,100 rise with the old 1,600 run would be 62.7 deg. The 2,200 run gives 54.6 deg. A longer flight and landing must be modelled, not a stretched steep ladder.

04 Fire and ventilation

New machinery volume affects fire suppression and ventilation. Recalculate protected volumes, fan duties and exhaust support. Do not reuse the old 190 m3 CO2 basis.

05 Pipe routing is reopened

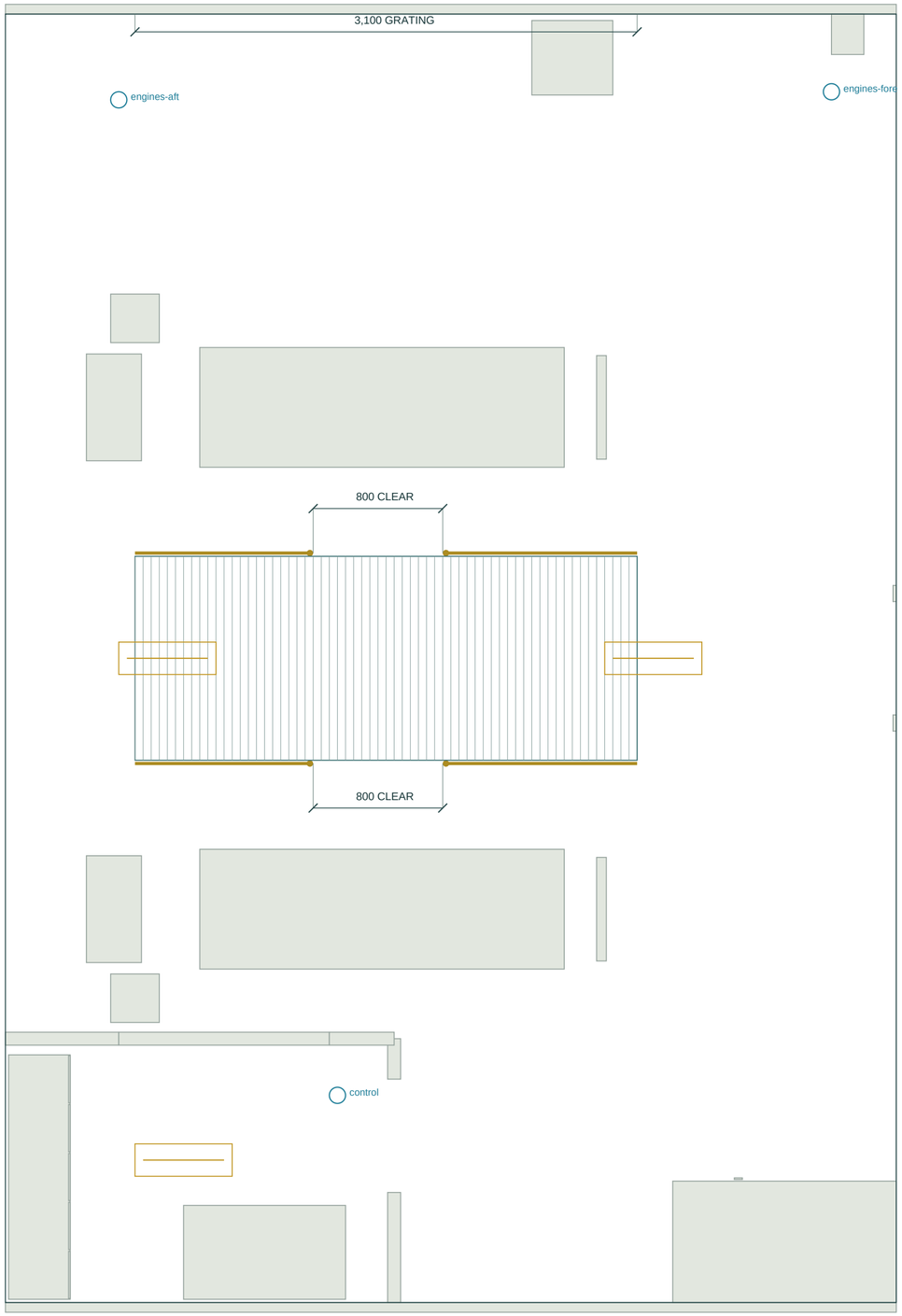
Baseline accepted paths are not accepted against H1 headroom. Move route design to the new overhead zone and recheck ownership, valves, supports, penetrations and escape.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

MINATO MARU / ENGINE ROOM H1

JS127-ER-H1-03 | CAD implementation amendment | 31 October 2026

Development geometry study. Source positions retained; dimensions in millimetres. Plan 1:25 at A2.

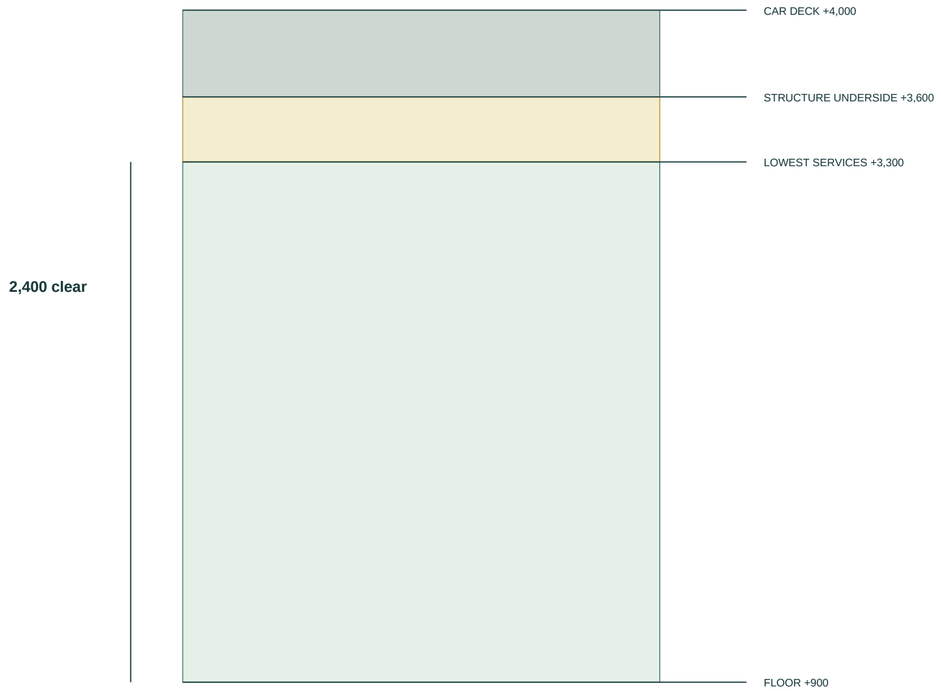


PLAN / machinery floor +900 | port up | X from transom

Grating 3,100 x 1,260; 6 mm bars at 50 mm pitch; top +900.

Shafts, engines, gearboxes and tank-top levels retained.

HEIGHT AND SERVICE ACCESS / 1:20



CHECKED IMPLEMENTATION DETAILS

Rail posts: X10,380 and X11,220; diameter 40. Inner faces leave 800 clear.

Both inboard engine rails have the service opening. Outside rails remain.

Doors X10,900 / 14,000 / 19,000 / 22,500: 2,100 high; widths 700/800/750/700.

Five new luminaire reservations: 600 x 200 x 50; underside +3,400.

Four retained fittings: two 300-square main-room; two 1,200 x 240 electrical.

Fixture centres and all eight fixed cameras are in design-h1.json and DXF.

All fixed lens centres: finite and clear by radius 120 against source boxes.

Working corridor: 1,260 between 40 mm inboard rails; floor top +900.

Clear headroom describes the working routes, not the occupied engine volume.

Illustrative pipe previews are omitted; complete service routing remains open.

400 mm deck zone is a reservation, not calculated structural scantlings.

Structure, stability, berth, fire volume, emergency escape and OEM lifts: OPEN.

Blender and World use shared geometry. Meshy pass pending human verification.

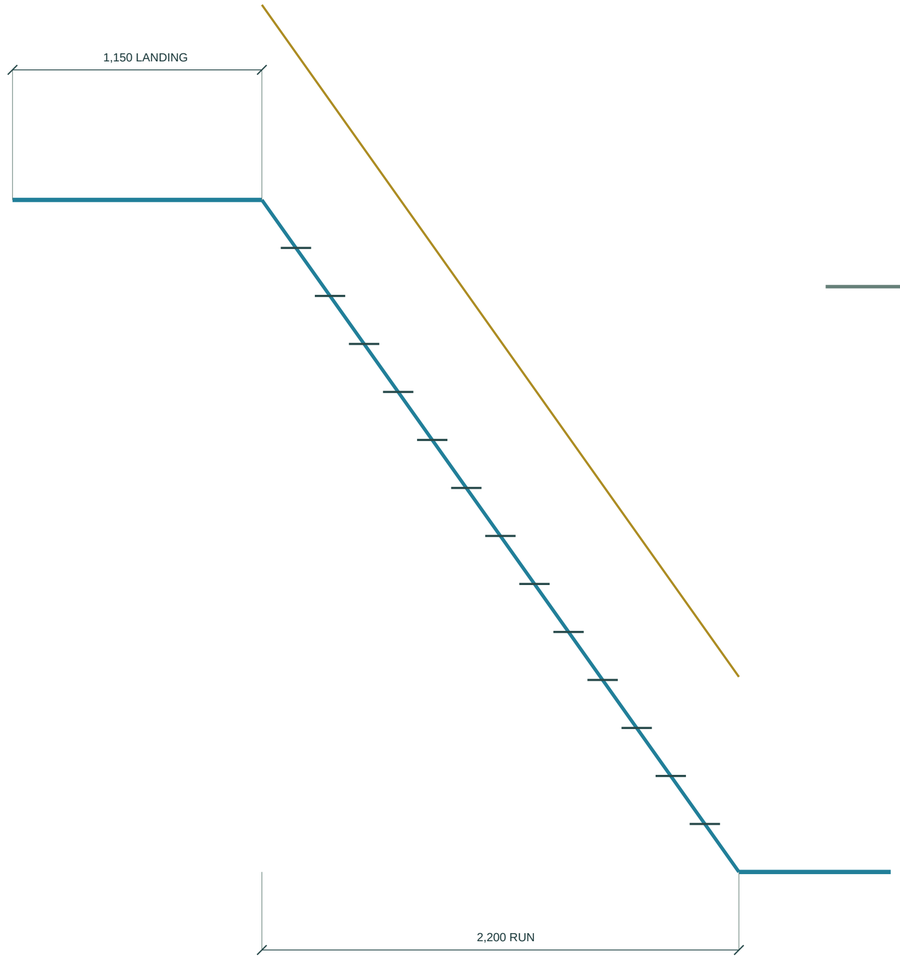
CAD audit: mm units; native dimensions; editable layers; model comparison.

H1 is an opt-in development study. No main promotion or fabrication approval.

MINATO MARU / H1 LADDER AND TOP LANDING

JS127-ER-H1-04 | Corrected access detail | 1:20 at A2 | mm

The longer ladder requires a landing behind its head. Relocate the casing P side door; do not enter mid-flight.



LONGITUDINAL SECTION / X forward from transom; Z above keel

Main access only. No protected escape or lifting-route acceptance is implied.

Development geometry study; not fabrication or regulatory approval.

ACCESS AUDIT / CLOSED FOR THE PROTOTYPE

New side-door opening X5,300-6,400; clear longitudinal opening 1,100.

Top landing X5,250-6,400: 1,150 long, above the same +4,000 car deck.

Ladder head X6,400; foot X8,600; rise 3,100; nominal angle 54.64 degrees.

14 risers at 221.43; nominal going 157.14. Flight is an inclined ladder.

Rail centres Y3,130 and Y3,970; diameter 40: 800 between inner faces.

Tread span Y3,140-3,960: 820 nominal. Hatch Y3,150-4,130: 980.

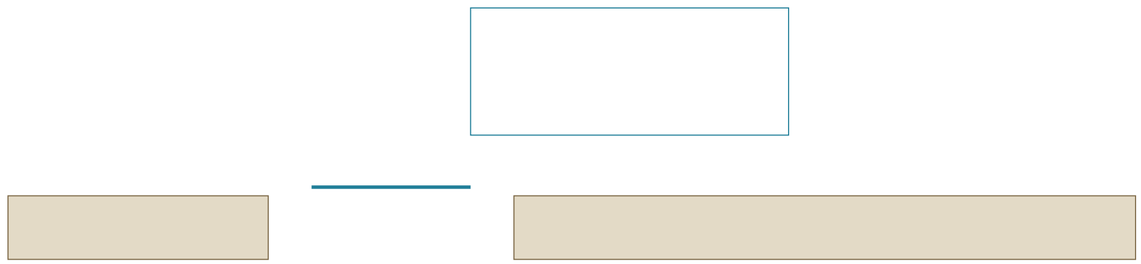
Open car-deck hatch X6,400-9,000. Ladder landing is not a second ER floor.

Walk down and back up using the new door and landing, not old X6,900.

Keep existing machinery, floor +900, tank top +700 and shaft axes.

Whole-vessel and protected emergency escape design remain open.

PORT LOBBY / DOOR AND BENCHES / 1:30



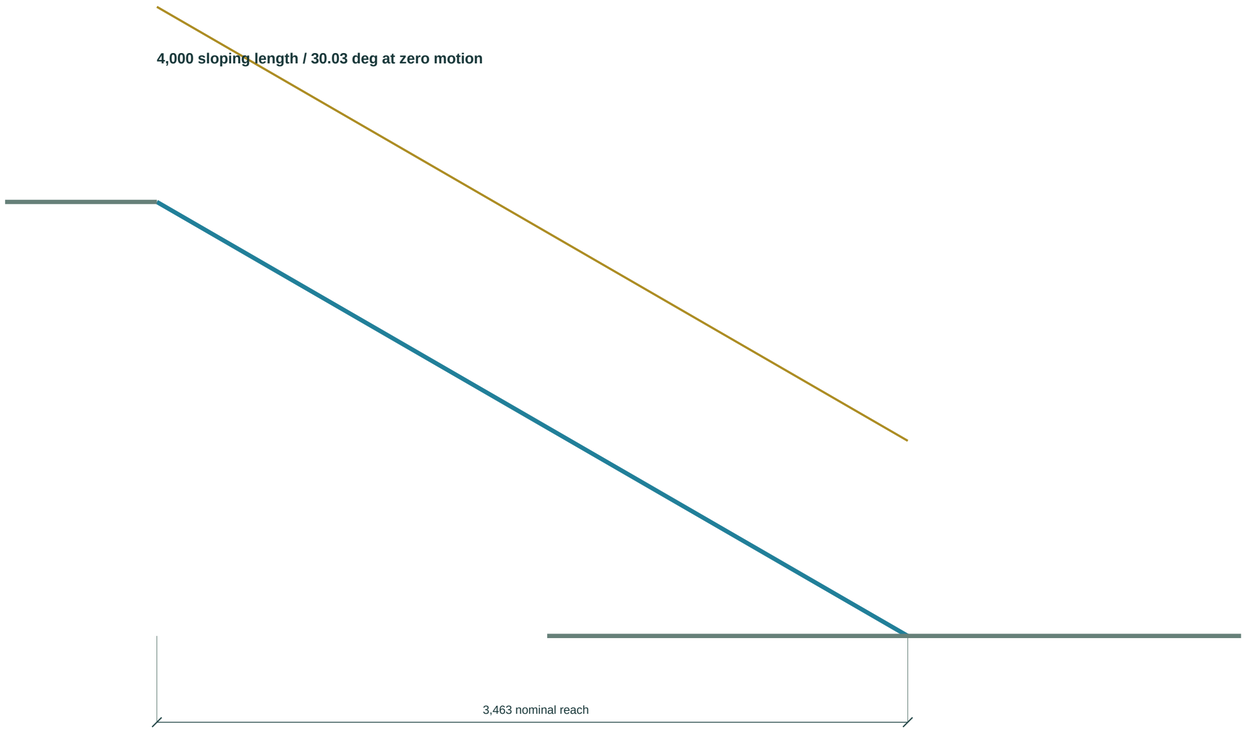
Door X5,300-6,400. Benches X3,200-5,000 / X6,700-11,000.

Retain total bench length 6,100; redistribute 1,000 forward.

MINATO MARU / H1 BOARDING COORDINATION

JS127-ER-H1-05 | Gangway and crew station | 1:20 at A2 | mm

Raised vehicle deck: extend the boarding gangway rather than leave an unsupported step from the pier.



COORDINATED GEOMETRY / LIMITS

Gangway: 4,000 long x 1,400 gross width; retained side handrails.

Fold upright under passenger deck; 4,200 clear zone reserves 200 margin.

New length maintains about 30-degree nominal boarding inclination.

Pier landing, clear pedestrian route, toe contact: verify in the walk tests.

Forward mooring station follows actual forecastle floor +6,460 above keel.

Crew post height derives from the bow-house top +60, not old 3.56 local.

Bow ramp geometry retained; both nominal berth routes require walk checks.

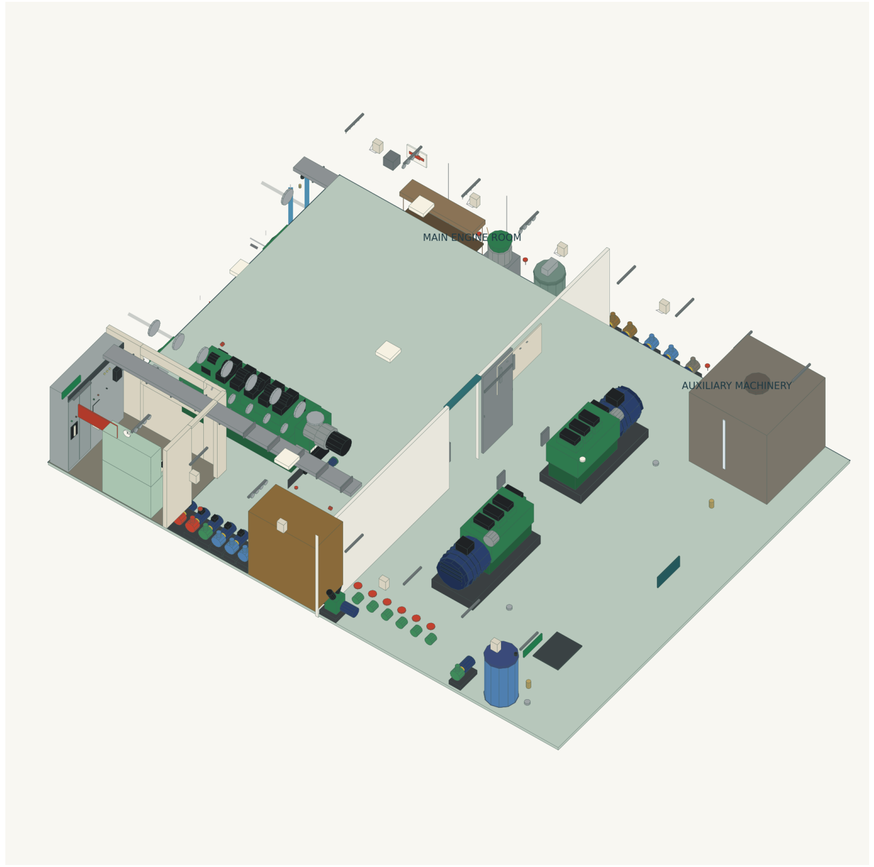
Hoist capacity, stow restraints, structure, tides and real access: OPEN.

No production release is authorised by this drawing.

SECTION / waterline datum uses draft 1,900; existing Minato pier top +98.

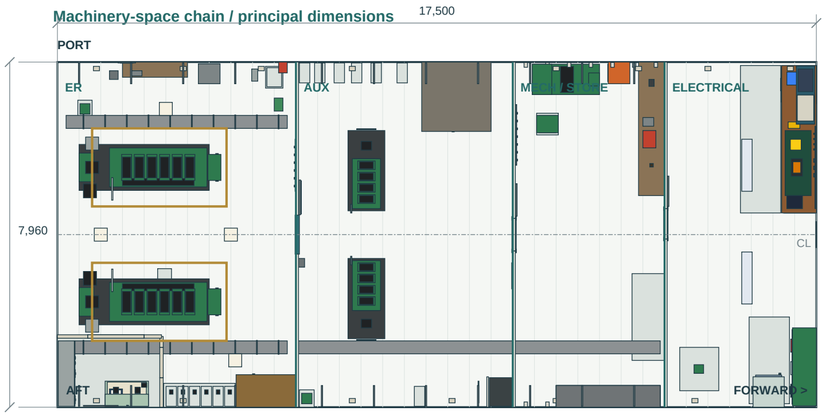
Runtime solves the actual top-surface contact under ship pitch and roll.

This is nominal game-geometry coordination; tidal and hoist design remain open.



DECKHEADS AND EXTERNAL WALLS OMITTED FOR VISIBILITY ONLY

Machinery-space chain / principal dimensions



01 Existing ferry retained

34,500 LOA x 8,400 moulded breadth. Twin-screw, bow-ramp island ferry. This issue draws current machinery; no equipment is added, deleted or relocated.

02 Propulsion retained

Two fictional Johansson Marine JM6-160 six-cylinder diesels: 460 kW at 1,350 rpm each; reverse-reduction gears aft, 3:1. Engines point forward; shafts leave aft.

03 Power and auxiliaries

Two 100 kW diesel-generator sets in the auxiliary room. Fire/GS, bilge, fuel transfer, purifier, air receivers, compressors and domestic systems remain in place.

04 What realism requires

Preserve correct scale, connected services, access to every maintainable part, protected exits and a practical lifting route. Simplify mesh detail only after these physical relationships work.

CURRENT MODEL SURVEY + ENGINEERING REVIEW

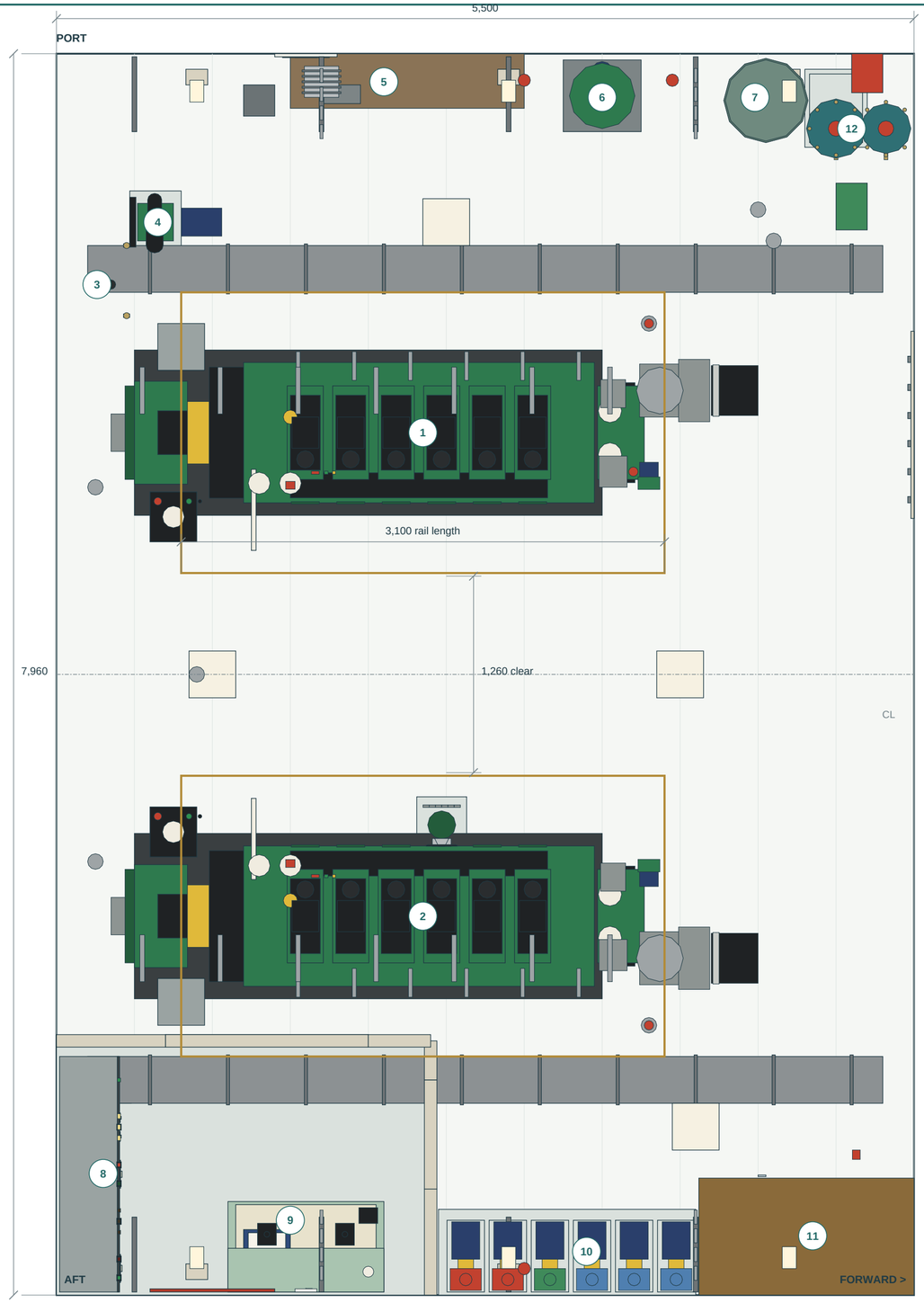
Dimensioned from main f49f22ba82e6 | 31 October 2026 | Units mm; dimensions govern.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

Drawn: source-derived | Checked: computational / visual | Approval: none

Main engine room / equipment and working clearances

REV C | 1:25 @ A2



1/2 Main engines P / S

Current bedplates X=9,000-12,000; centres Y=+/-1,550. Actual fittings are projected from source; manufacturer installation dimensions remain unverified.

3/4 Starting-air equipment

Two 150 L receivers and compressor No.1 aft to port. Receiver reliefs, drains, certified start capacity and complete piping still need detail.

5-7 Port service bench / treatment

Workbench, fuel purifier and oily-water separator. Keep machinery opening and filter/basket withdrawal space; the picture alone does not prove service access.

8/9 Main switchboard / control booth

MSB aft, engine console and log desk along the side. Current booth has one modelled doorway; protected independent escape is unresolved.

10 Starboard pump row

Fire / GS No.1; fire / bilge No.2; GS; bilge; sanitary No.1; hydrophore No.1. Sequence and installed count retained.

11/12 Fuel service tank / sea-water strainer

Service tank remains forward starboard. Both sea-water suctions/strainers must remain accessible. Tank, hot exhaust and switchboard routes must be coordinated.

Closed yellow rail loops currently surround both engines. No gate is modelled; a maintainable machine cannot rely on climbing a rail. A correction must retain guarded edges and provide deliberate service access.

CURRENT MODEL SURVEY + ENGINEERING REVIEW

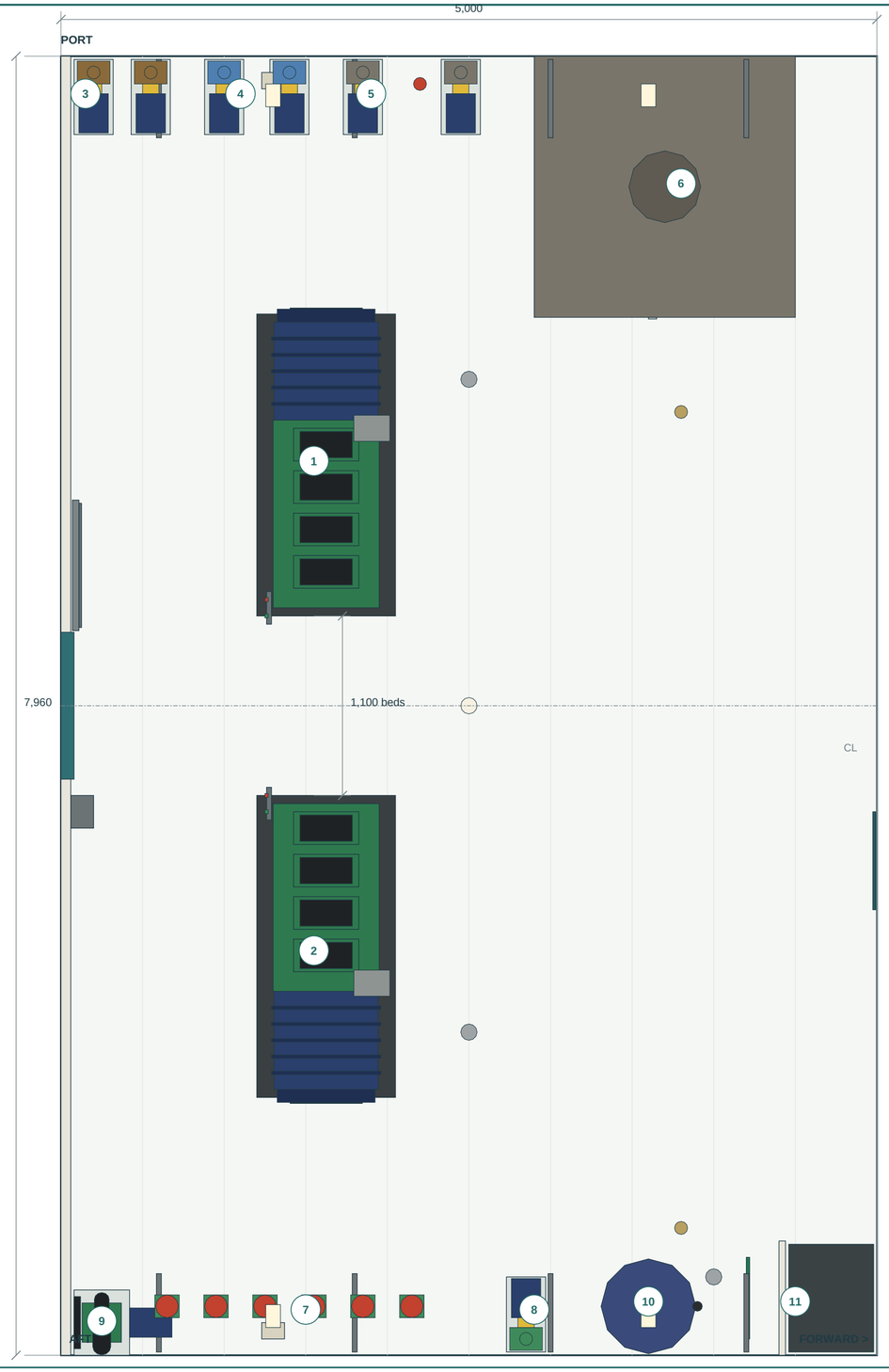
Dimensioned from main f49f22ba82e6 | 31 October 2026 | Units mm; dimensions govern.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

Drawn: source-derived | Checked: computational / visual | Approval: none

Auxiliary machinery room / generators and service plant

REV C | 1:25 @ A2



1/2 Diesel generators No.1 / No.2

Current set beds X=15,200-16,050, Y=550-2,400 and -2,400 to -550. Inboard dynamic fittings leave a smaller declared 1,000 mm working aisle.

3 Fuel transfer No.1 / No.2

Two separate pumps remain. Dedicated suction/discharge interfaces and control/bypass arrangements need the unpublished fuel C1 design to be reviewed.

4/5 Domestic standby / sewage pumps

Hydrophore No.2, sanitary No.2 and sewage discharge No.2. Sewage discharge No.1 is between the pump row and holding tank.

6 Sewage holding tank

Nominal 4 m3, current external envelope 1,600 x 1,600 x 1,700. Gross envelope is 4.352 m3, before platework, internals and unusable volume.

7-10 Starboard services

Ballast manifold, ballast pump, compressor No.2 and 300 L hydrophore vessel. Access to valve spindles, motor withdrawals and pressure-vessel inspections remains a hold.

11 Forward starboard escape

A 700 mm-wide model trunk is drawn. It is an existing representation, not a proved protected exit to a safe deck. Route and fire-boundary details must be completed.

Door at X=14,000: 800 mm nominal wide. Door at X=19,000: 750 mm nominal wide. Do not route machinery removal through these openings unless the dismantled item and rigging fit.

CURRENT MODEL SURVEY + ENGINEERING REVIEW

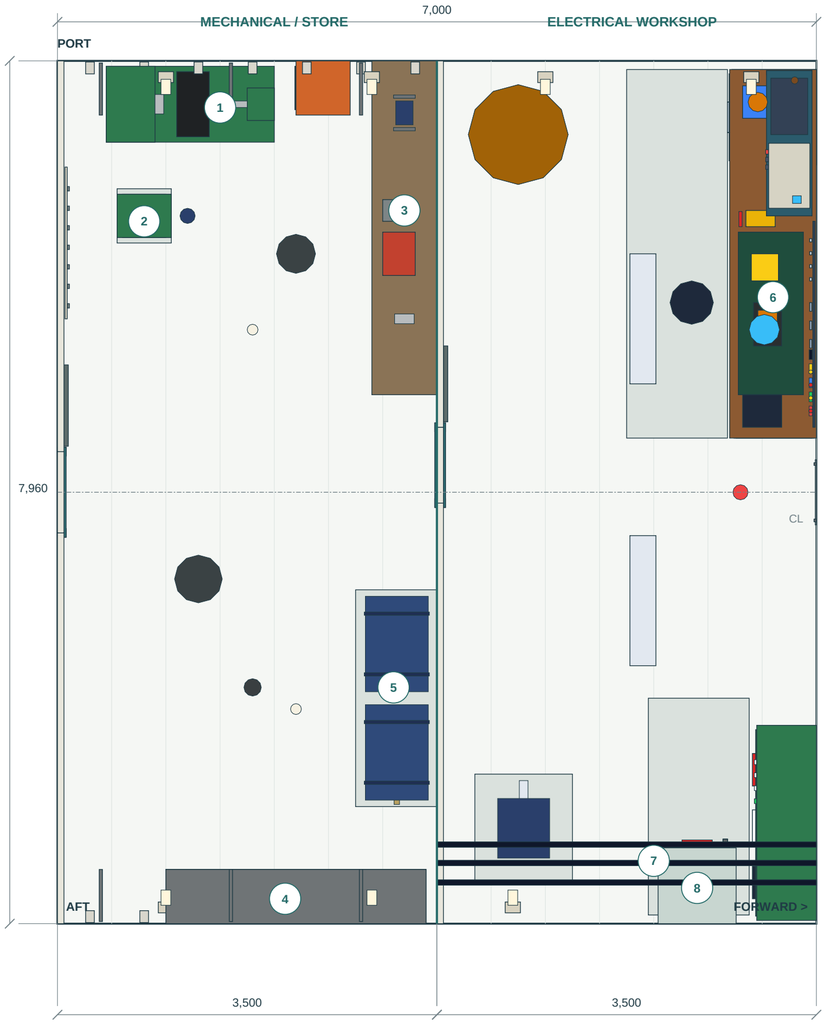
Dimensioned from main f49f22ba82e6 | 31 October 2026 | Units mm; dimensions govern.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

Drawn: source-derived | Checked: computational / visual | Approval: none

Mechanical workshop, engine store and electrical workshop

REV C | 1:40 @ A2



1-3 Mechanical work

Lathe, pedestal drill, main bench and vice remain. Welding set and grinder remain in the room; hot-work control, screens and extraction are not established by the model.

4/5 Engine store

Spares racks, two oil drums and chain block remain. Mesh partition retained. A stored chain block is not a designed lifting beam or a complete engine-removal scheme.

6/7 Electrical work

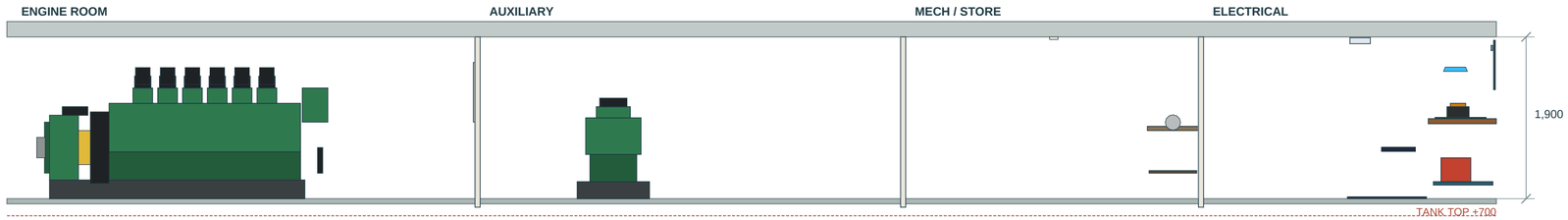
The later electrical workshop is included, unlike the 13 October Rev B source snapshot: instrument bench, test panel, insulated mat and motor service area.

8 Electrical escape

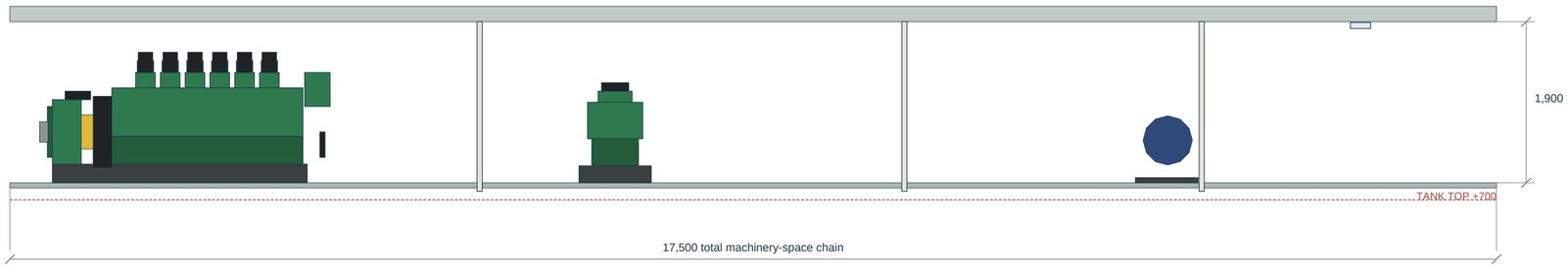
Existing trunk at X=24,600-25,200 remains. Its through-deck continuation and hatch still need detailing. Do not treat the store as a protected safe escape location.

Current plan only. No machinery, furnishings or compartment boundaries moved. Workshop/store subdivision is the game source; its real damage-stability and fire-division basis is unresolved.

A-A / port machinery centre plane, Y=+1,550



B-B / starboard machinery centre plane, Y=-1,550



Double-bottom liquid volumes are existing allocations only: two 7.5 m3 fuel-storage tanks; bilge 1.0 m3; sludge 0.3 m3; lube oil 0.6 m3. No individual tank partitions, calibrated capacities or fabrication scantlings are invented here.

The source hull and loading assumptions remain unreconciled. Do not lower the machinery floor, deepen tanks or move bulkheads to manufacture clearance without rechecking hull shape, structure, weights and stability.

Vertical geometry governs

01 Floor / tank top

Tank top +700; floor plates +900; deckhead +2,800; car deck +3,000. Nominal headroom 1,900 before cable trays, pipes, stiffeners and insulation.

02 No imaginary extra storey

The source grating sits at about +906, on the working-floor level. The 200 mm tank-top interval is service depth, not a walkable lower machinery deck.

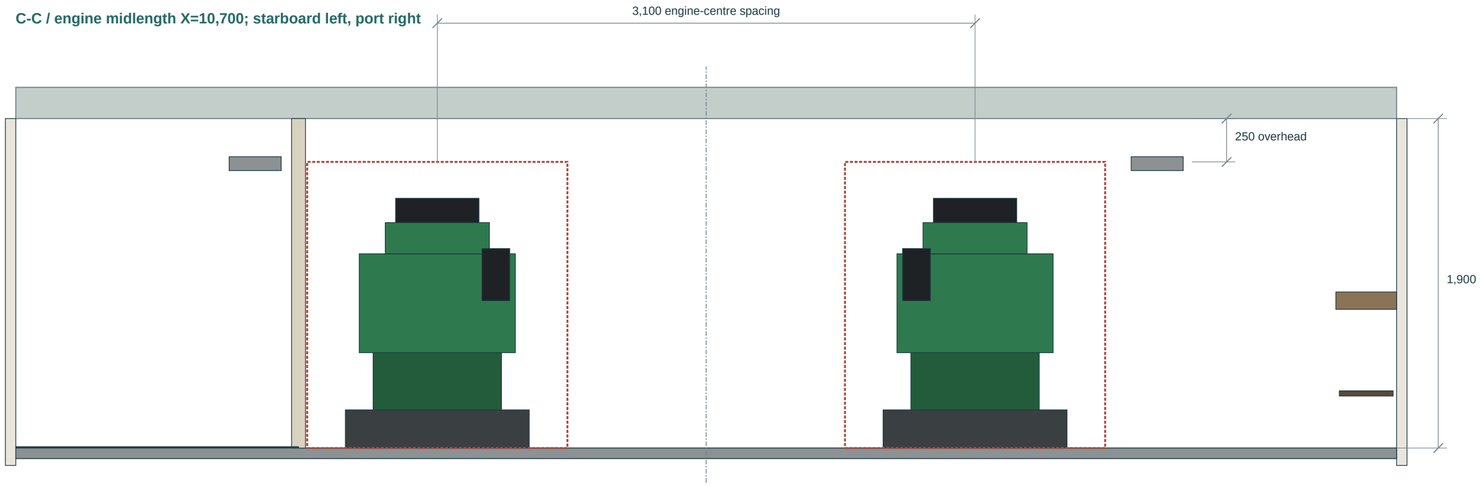
03 Overhaul above each engine

The M1 reference reservation rises to +2,550, leaving 250 mm to the deckhead. Piston/liner withdrawal and chain-block hook height are unknown.

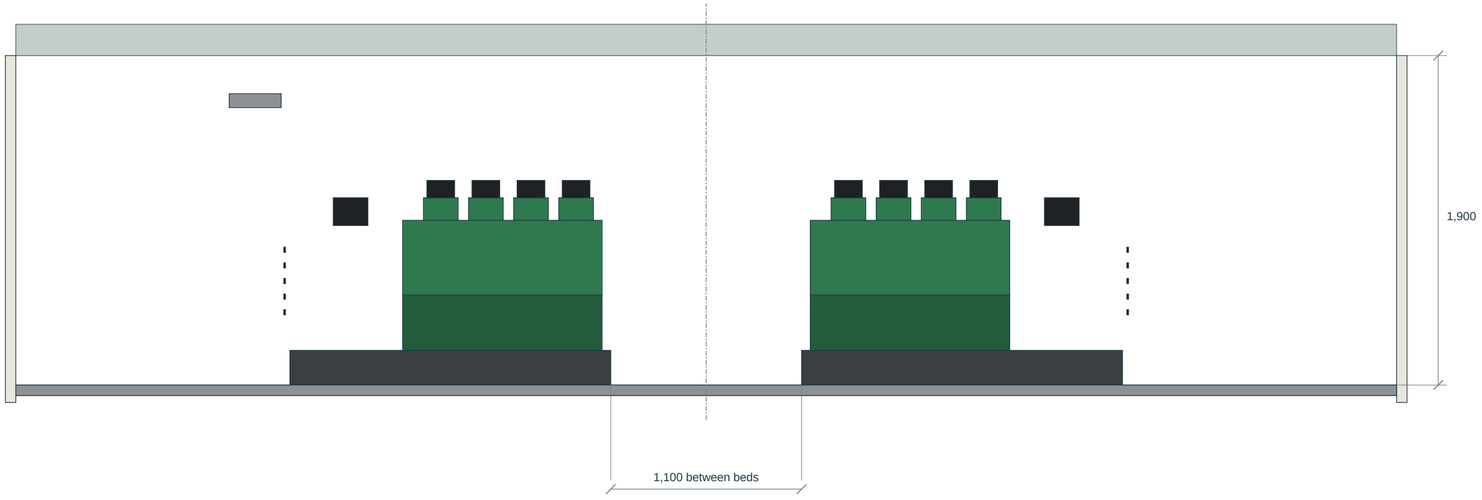
04 Structural space

Deck beams, tank-top girders, foundation bolts, resilient mounts and lifting points are not fully engineered. The 180 mm model deckhead is an envelope, not an approved beam design.

C-C / engine midlength X=10,700; starboard left, port right



D-D / generator station X=15,600



01 Measured versus reserved

Solid colour is the current game model. Red dashed boxes are retained M1 engine/gear reservations: 2,700 x 1,500 x 1,650, not manufacturer installation data.

02 Watchkeeper route

A 2,000 mm clear route-height target cannot fit inside the existing 1,900 mm nominal room. Actual piping/trays reduce this further. Do not hide the conflict with a smaller avatar.

03 Aisles

Centre aisle is 1,260 mm between actual rail faces. Side access must be checked against fittings, control-booth glazing, valve swing and required part withdrawal, not just a body capsule.

04 Working positions

Model a standing operator, kneeling filter service, removal of a rocker cover and the largest planned spare. Keep those envelopes clear before adding decorative props.

CURRENT MODEL SURVEY + ENGINEERING REVIEW

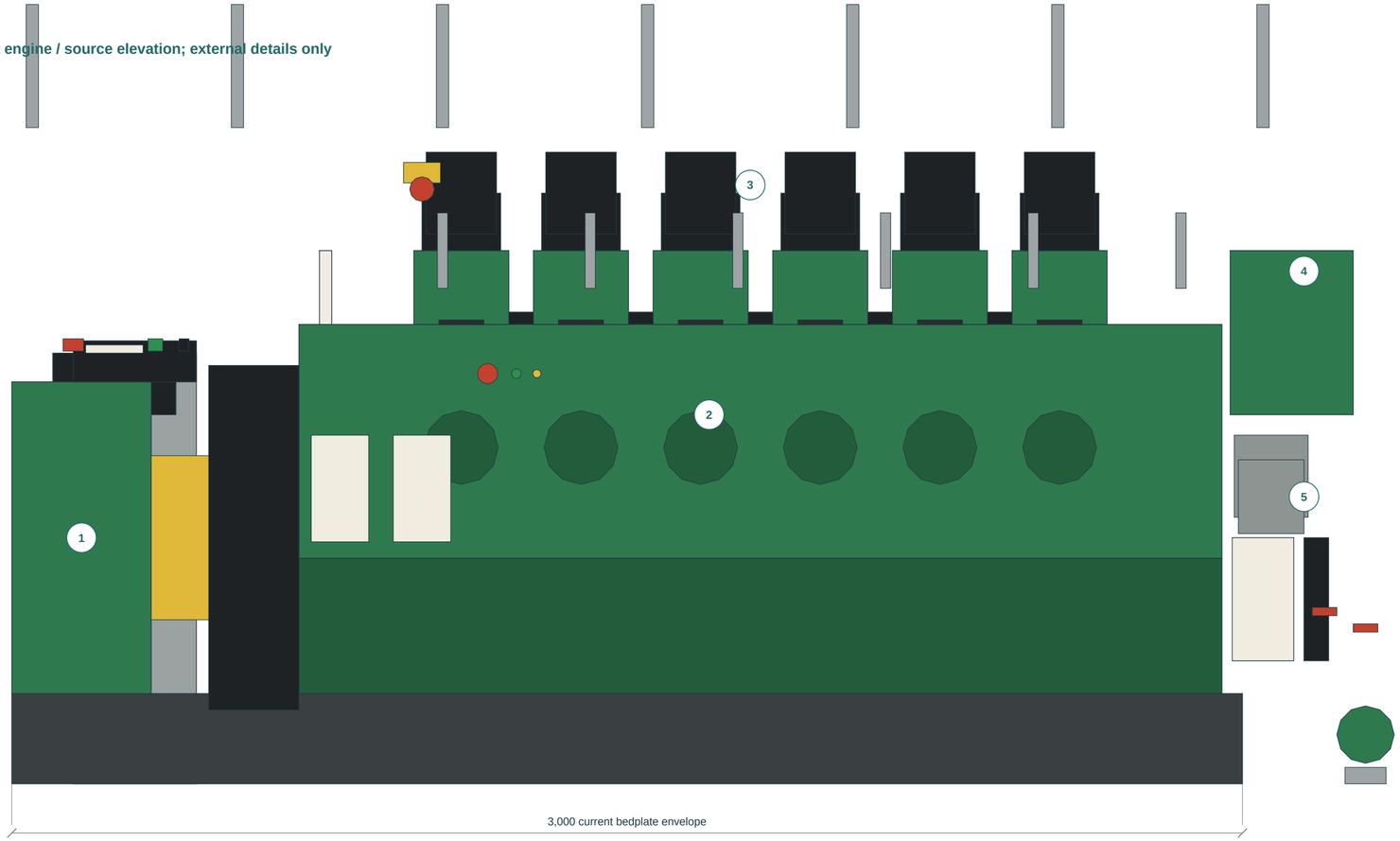
Dimensioned from main f49f22ba82e6 | 31 October 2026 | Units mm; dimensions govern.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

Main engine / recognisable machinery and controlled interfaces

REV C | 1:10 @ A2

Port engine / source elevation; external details only



AFT / GEARBOX + SHAFT <

FORWARD / TURBO >

1 Reverse-reduction gearbox

Gear aft, flywheel/coupling between block and gear; propeller shaft leaves aft. Confirm output height, thrust reaction and coupling guard from selected equipment.

2/3 Six-cylinder mechanical diesel

Six rocker covers, crankcase service doors, injection equipment and lagged exhaust must read as one coherent machine. A permanently open crankcase cover is inappropriate for the running state.

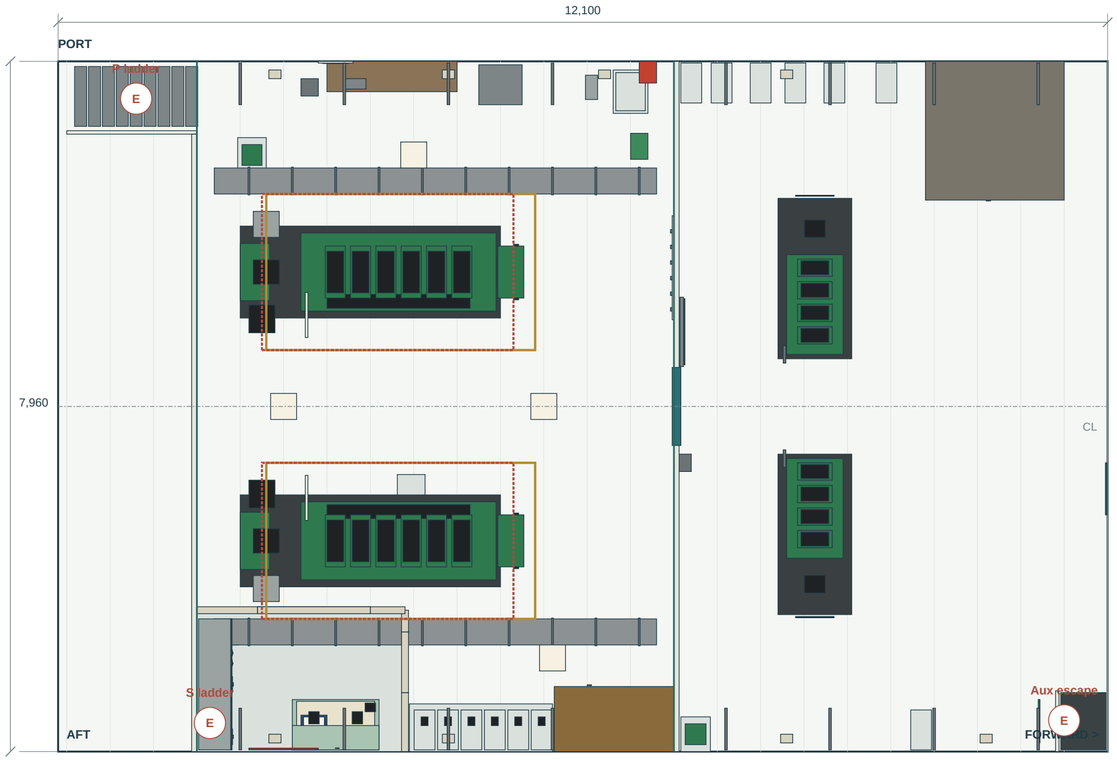
4/5 Turbocharger and cooling

Turbocharger forward, charge-air cooler and engine-mounted pumps; starting-air, fuel feed/return, LO and cooling connections must each have a separate functional interface.

Real machinery comparison / not an engine substitution

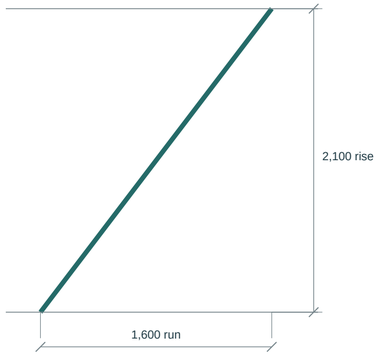
The retained JM6-160 is fictional: 460 kW / 1,350 rpm / 160 x 200 mm / six cylinders / 24.127 L. A verified Yanmar 6AYM-WST reference is 485 kW at 1,900 rpm, 155 x 180 mm and 20.379 L. The legacy outline is 2,000 x 1,305 x 1,431 mm engine-only; a YX181 package has 2,364 mm combined length. Do not mix those ratings, masses, starting systems or installation dimensions.

Drawing decision: retain installed count, purpose and shaft orientation. Obtain a period-credible matched engine/gear installation basis before replacing the fictional envelope with a detailed production asset.



Red dashed rectangles: earlier 2,900 x 1,800 removal-cover reservations; NOT existing deck openings.

Port ladder / actual rise and run



52.7 deg nominal inclination

750 mm ladder width. This is a geometric screen only; hands, hatch, head clearance and protected route are unresolved.

01 Escape route continuity

Trace both exits from the lowest occupied level to an unobstructed safe deck, including closed-door states. Current aft S ladder has 400 mm rung span, but no complete protected trunk is established.

02 Control booth and workshops

The present booth has one doorway back into the machinery space. Review an independent sheltered way out; do not count a view through glazing, an oil store or a locked room as an exit.

03 Human reference, correctly used

ClassNK Part R Chapter 13 is a design-check reference, not an asserted rule approval for this fictional 1994 domestic ferry. It distinguishes ladders from 800 x 800 clear vertical shelter envelopes and 600 mm horizontal shelters.

04 Engine removal

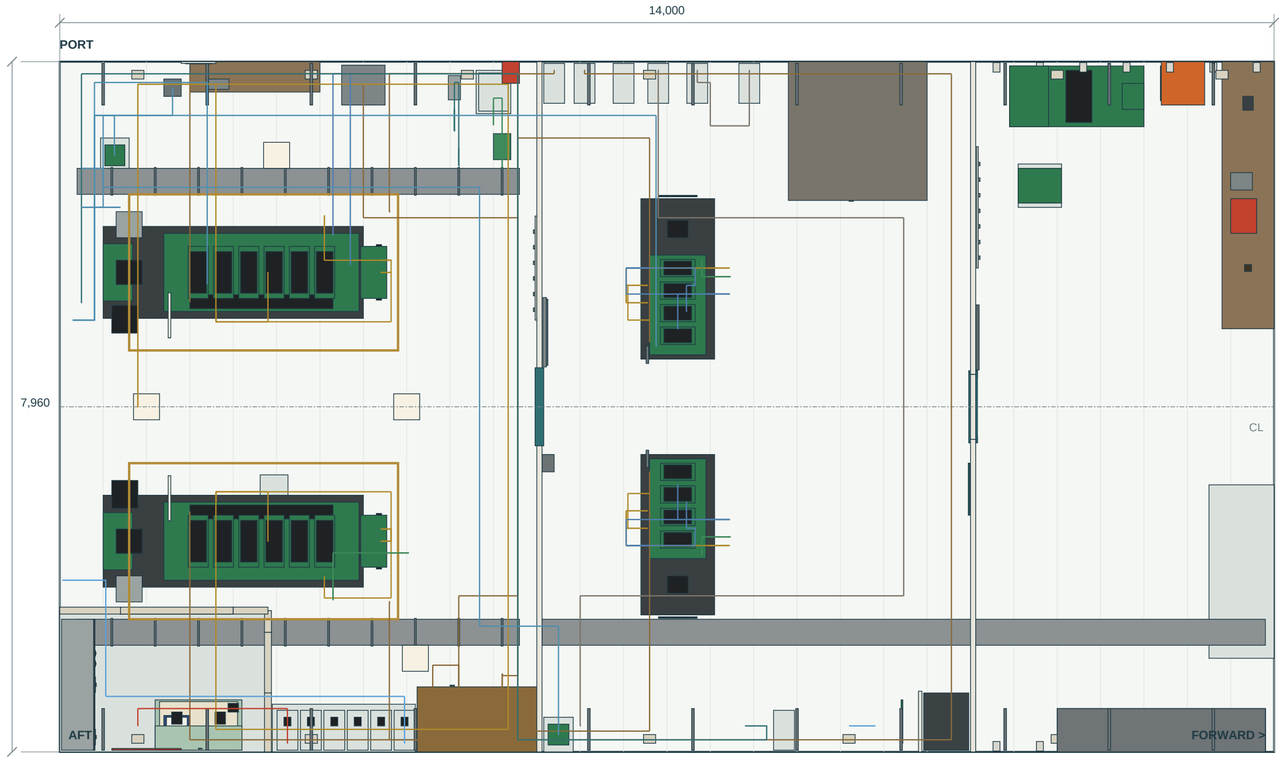
The prior 3.6 t machinery allowance plus 0.4 t rigging is a planning 4.0 t lift only. No beam, padeye, hook-height, open-deck crane path or through-car-deck closure has been proved.

05 Do not cut the car deck yet

Reserve removal openings only after coordinating vehicle loading, deck girders, watertight/weather closures, hatch handling, shore crane reach and lift stability.

Services coordination / geometry-screened routes and open connections

REV C | 1:50 @ A2



Only the 55 current geometry-screened routes are shown. Their nozzle definitions and fabrication design remain unverified.

Service	Graph lines	Screened	Open physical
Fuel oil	22	10	12
Lubricating oil	28	16	10
Sea-water cooling	23	4	19
Jacket water	24	9	15
Bilge and ballast	24	4	20
Fire main and CO2	22	1	19
Starting and service air	17	6	11
Fresh water	15	2	13
Sanitary water and sewage	10	3	7
Exhaust gas	8	0	8
Hydraulics: steering and ramp	17	0	17

01 Current main baseline

206 positive-bore physical connections: 55 geometry-screened and 151 unresolved. Four further graph edges are signals. The 172 coloured game preview paths are excluded from the engineering-route overlay.

02 Fuel C1 already exists

PR #192, head 9761f61304d3, is OPEN and unmerged. It proposes distinct fuel interfaces and reports 67 screened / 139 unresolved whole-vessel connections. Those are candidate results, not current main.

03 Fuel head contradiction

Current main still states 1.8 m gravity head. Model service outlet Z=2,200, engine feed Z=2,050: only 150 mm interface elevation difference, not tank operating head. C1 already removes the unsupported assertion.

04 Route every function

For each pipe: identify origin, destination, bore/class, flow direction, valves, drains/vents, supports and bulkhead sleeve. Preserve maintenance and escape envelopes; a coloured line alone is insufficient.

05 Ventilation / exhaust / fire

Prove combustion and heat-rejection airflow with ducts and dampers; thermally separate exhaust from oil systems; define protected volumes and extinguishing boundaries before sizing or adding detail.

CURRENT MODEL SURVEY + ENGINEERING REVIEW

Dimensioned from main f49f22ba82e6 | 31 October 2026 | Units mm; dimensions govern.

FOR LAYOUT REVIEW. GAME UNCHANGED. OPEN ENGINEERING ITEMS ARE SHOWN; NO FABRICATION RELEASE.

Drawn: source-derived | Checked: computational / visual | Approval: none

Disposition: measured baseline complete; physical design remains open

The drawing set preserves the current ferry. It is a concrete basis for correction, not evidence that the existing engine room is already fully realistic. The following gates must close before new equipment, demolition or a replacement interior is commissioned.

ID	Review item	Existing evidence	Correction / acceptance gate
R01	Headroom and overhaul	1,900 nominal; 250 above M1 envelope.	Choose viable maintenance/lifting method, including cover and hook travel.
R02	Access around engines	Closed rail loops; side/booth interactions.	Design service gates and verify personnel / largest-part envelopes.
R03	Escape and enclosure	One booth door; incomplete escape trunks.	Draw continuous, separated exits and fire boundaries through to safe deck.
R04	Equipment identity	Fictional engine and machinery outlines.	Keep fictional branding; fix one coherent period-credible installation basis.
R05	Pipes and interfaces	151 main connections unresolved.	Review existing fuel C1; coordinate other services with real valve/nozzle space.
R06	Foundations and deck loads	No complete beams, mounts or lift structure.	Engineer supporting steel and shaft reactions before any deck opening.
R07	Tanks / bilges / flooding	Shared reservations; individual partitions open.	Define boundaries, suctions, vents, access, tank calibration and bilge wells.
R08	Air / heat / fire	Fan flat unbuilt; room boundaries incomplete.	Coordinate ventilation, cooling, exhaust lagging and suppression coverage.
R09	Power and control	Electrical ratings are preliminary.	Verify motor starts, protection, cable routes and black-out / emergency supply.
R10	Whole-vessel fit	Hull / weight assumptions unreconciled.	Check hull, stability and subdivision before changing floor or compartment size.

Source and reference control

Model: nj22az/nj22az.github.io main f49f22ba82e6d4b1fd9b5033f8bddc4d3eed903f; source modules and data are frozen in this package.

Prior baseline: Minato Maru Engine Room CAD Rev B, 13 October 2026; retained unchanged as a separate reference.

Fuel candidate: https://github.com/nj22az/nj22az.github.io/pull/192 | 9761f61304d3ac7a66d9061b593d61fce706805c | unmerged.

Yanmar: https://www.yanmar.com/media/news/2020/05/22051608/6aym-wstwet.pdf | operating reference; later than the setting.

Yanmar legacy outline: https://www.yanmar.no/theme/yanmarportal/uploadedFiles/Marine/productDownloads/Commercial-datasheet/6148-Datasheet-6AYM-WST.pdf

ClassNK: https://www.classnk.or.jp/hp/pdf/rules/460_part_r_e_202606.pdf | Ch.13 / R13; retrieved PDF labels its issue December 2025.

Release boundary: no game source, runtime, equipment, residents, hull or ship schedule changed. Current pipe counts were rerun from the source export. Historic Rev B and C1 test counts are not claimed as newly run tests.